

Introduction

Distributed Denial of Service attacks, or DDoS attacks are attempts to make digital infrastructure unavailable by flooding them with requests from multiple sources. A malicious user infects other devices, allowing the user to control them. When all infected devices flood the same system, the system breaks down, and ordinary traffic is unable to reach the desired system. One of the most famous DDoS attacks was in 2016; more than a million devices attacked DNS provider Dyn, resulting in major online services like Netflix, Airbnb, and the BBC being cut off. More recently, social media service Bluesky was attacked for 24 hours on April 15th by a pro-Iran hacktivist group.

Incidents like these show the relevance of protection and detection of DDoS attacks.

Distributed Denial of Service attacks are (one of the) most common network security threat experienced by Enterprises, Governments and Educational bodies, while the customers of Service Providers (e.g. ISPs and Data Centers) remain to be the number one target of DDoS attacks [arbor networks] while network-layer attacks target vital Internet structure [cloudflare report]. DDoS attacks are easy to execute, even without technical knowledge [netscout]. AI integrated attacks and increasing sophistication only prove that DDOS attacks are becoming more dangerous: A record of 31.4 terabytes per second attack has been stopped by Internet-service provider Cloudflare [cloudflare].

Due to the relevance of DDoS attacks, many researches have been conducted on detection, defending, and mitigating attacks. In this research, we investigate the severity of a DDoS attack on its societal impact, where we use Google Alerts as a source to identify DDoS attacks. These alerts are collected in a database, which has been annotated as “Arrest”, “Attack”, “Research”, and “Other” in prior research [Wim kamerman].

We want to see how several dimensions influence the social ‘popularity’ of a DDoS attack. This can be recorded in its mentions in the dataset. The dimensions we record, but are not limited to: Political, Economical, Legal, Environmental, Social , and Technical impact. A common analysis models which combines these factors, is the PESTLE analysis model, which we will use.

The main goals of this research are:

- Analyse annotated Google alert dataset with PESTLE analysis model
- Create valuation model based on analysis
- Expand current annotated Google alert dataset with analysis data
- A tool that can generate insight into the (analysed) impact of a DDoS attack given a dataset

Commented [JD1]: Is het raar om over de oorlog te beginnen als dat natuurlijk niet de aanleiding is?

- Research question

RQ1: What is the impact of PESTEL dimensions on the news popularity of a DDOS attack?

RQ2: How do we create a measurement tool to contextualize/visualize the impact (based on the annotated dataset)?

Methodology and approach

The dataset we want to use has already been collected. We use a dataset, comprised of Google Alerts, that has been described by Abhishta et al. [Abishta]. The dataset consists of Google Alerts on the keywords “DDoS” and “Distributed Denial of Service”. The dataset has been annotated as “Arrest”, “Attack”, “Research”, and “Other” in research by Wim Kamerman [Wim].

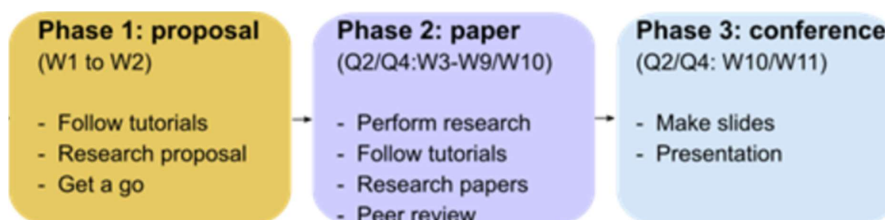
We perform a manual, qualitative analysis on relevant PESTLE dimensions occurring in the dataset. We note which dimension occurs in which Google Alert notification. We store this metric in the dataset. We will use Python to create an artifact capable of interpreting the dataset, based on Kamermans research, and we will use a local, provided by the University of Twente Large Language Model to help code this artifact, and we will expand the Machine Learning models for text annotation provided by Kamerman.

Work programme

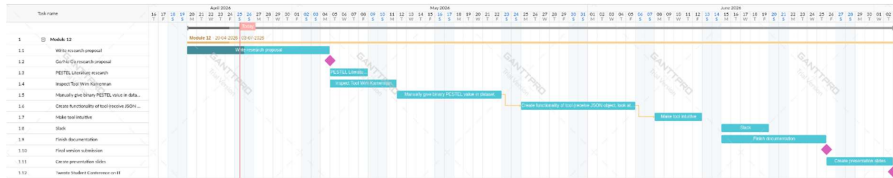
This module consists of 11 weeks. In this research, there are two substantial content-related tasks which need to be completed in the given timeframe:

- Give PESTEL value to dataset
- Create tool to gain insight into ‘value’ of alerts about DDOS

Next to that, we also need time for documentation (in-code and in the paper), preparing the presentation for the *Twente Student Conference on IT*, and we need time for setbacks. From the Students manual, we see:



This gives us the following schedule:



During the paper writing part, we divided the time as follows. The first week, time has to be made clear for setup and literature research. Since the research this paper is based on is from 2018, the libraries and code written are outdated, which will likely give some problems early on. Manually going through (a sample of) the dataset will be labour intensive. This process could be sped up using large language models, but a human viewpoint is probably preferred, which is why we think 2 weeks will be necessary to properly perform this task. Creating the tool, and making it intuitive and usable, will likely cost three weeks of time. A large part of this process can only be started once the dataset has been annotated, which is why it is essential we do the annotation correctly.

In the end, we have reserved 2 weeks of time for documenting purposes, as well as slack and review moments. It is likely that review moments are spread throughout the module; for simplification reasons we have grouped them together in the last weeks. In these last weeks, the conference needs to be prepared with slides, and the presentation must be practiced thoroughly. This will likely take another week. Together this research spans the entire module.